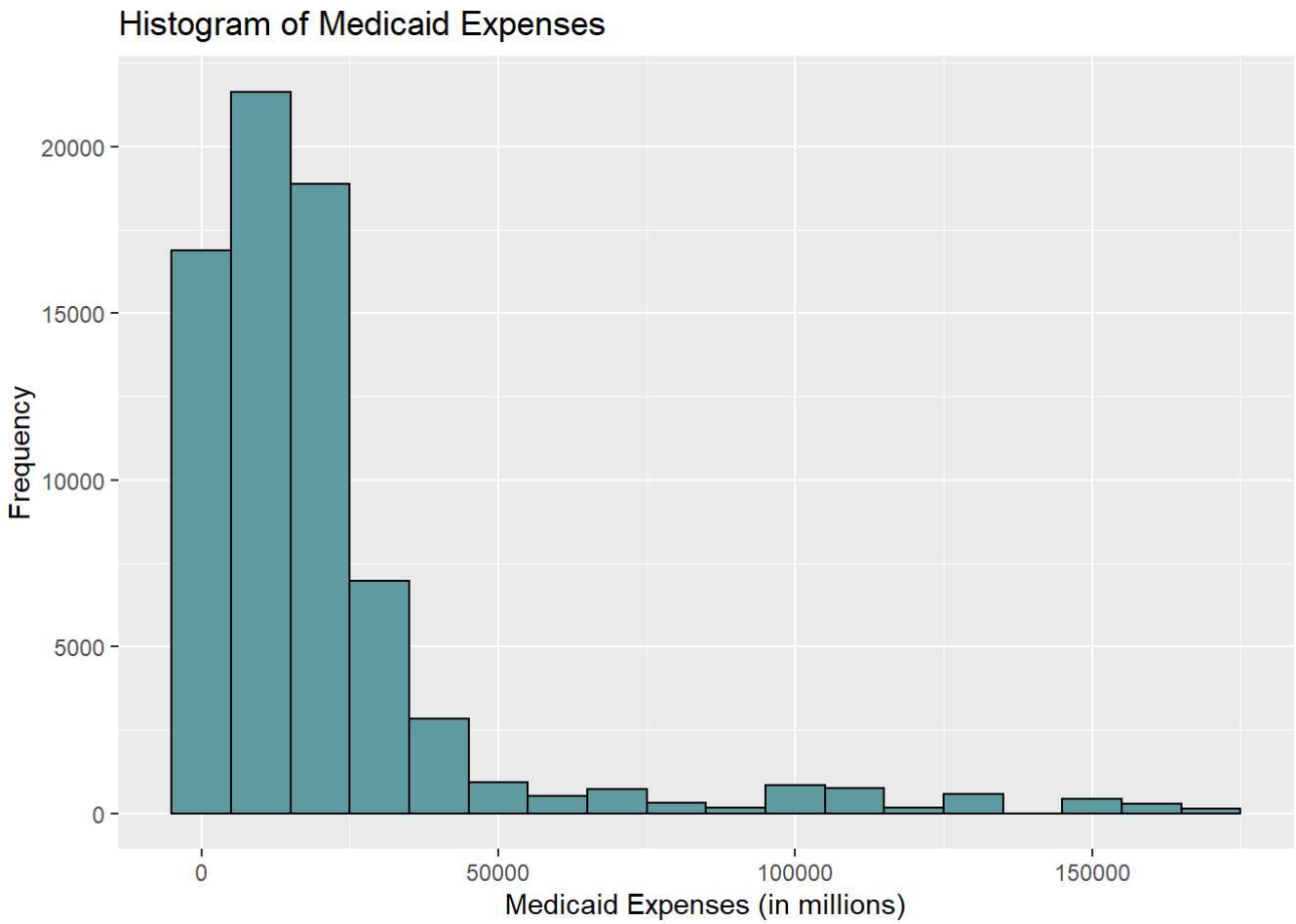
Chronic_Pulmonary_Analysis

2025-07-03

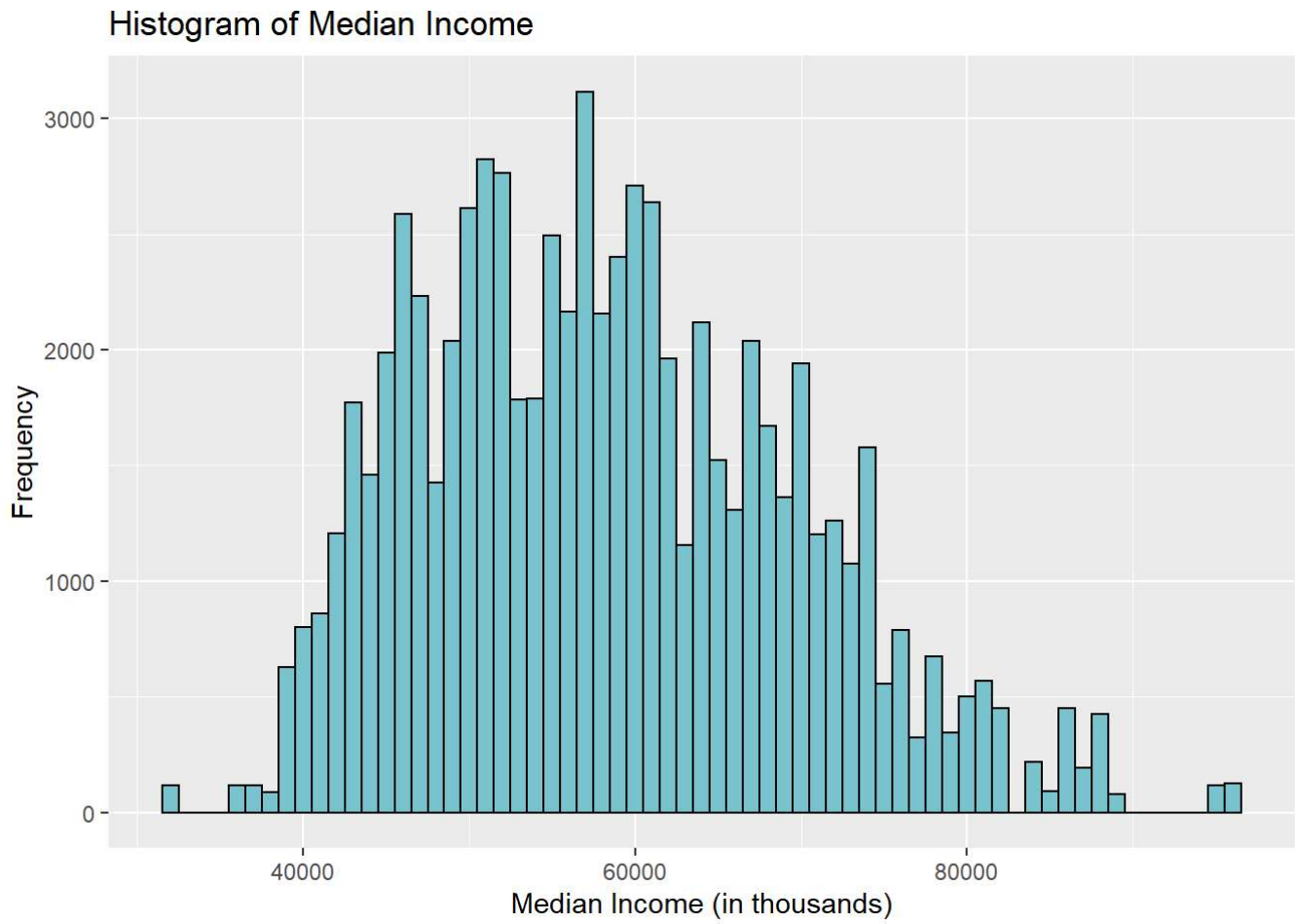
Medicaid Expenses Distribution

```
ggplot(final_df, aes(x = Medicaid_Expenses)) +
  geom_histogram(binwidth = 10000, fill = "cadetblue", color = "black") +
  labs(title = "Histogram of Medicaid Expenses", x = "Medicaid Expenses (in millions)", y = "Fre
quency")
```



Median Income Distribution

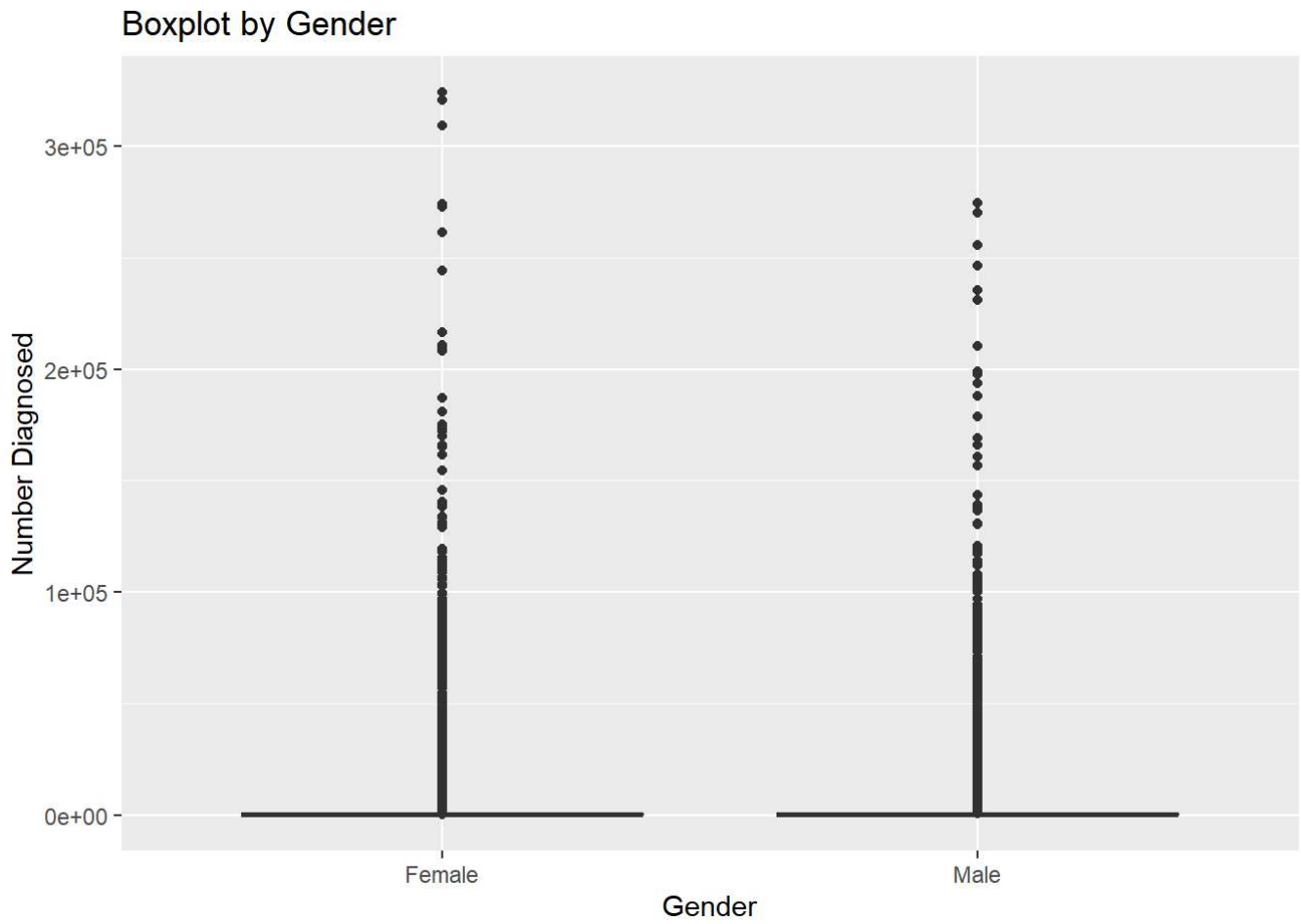
```
ggplot(final_df, aes(x = Median_Income)) +
  geom_histogram(binwidth = 1000, fill = "cadetblue3", color = "black") +
  labs(title = "Histogram of Median Income", x = "Median Income (in thousands)", y = "Frequenc
y")
```



Diagnosed Cases by Gender

```
# filter to gender
gender <- final_df %>% filter(StratificationCategory1 == "Gender")

# number diagnosed by gender
ggplot(gender, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Gender", x = "Gender", y = "Number Diagnosed")
```

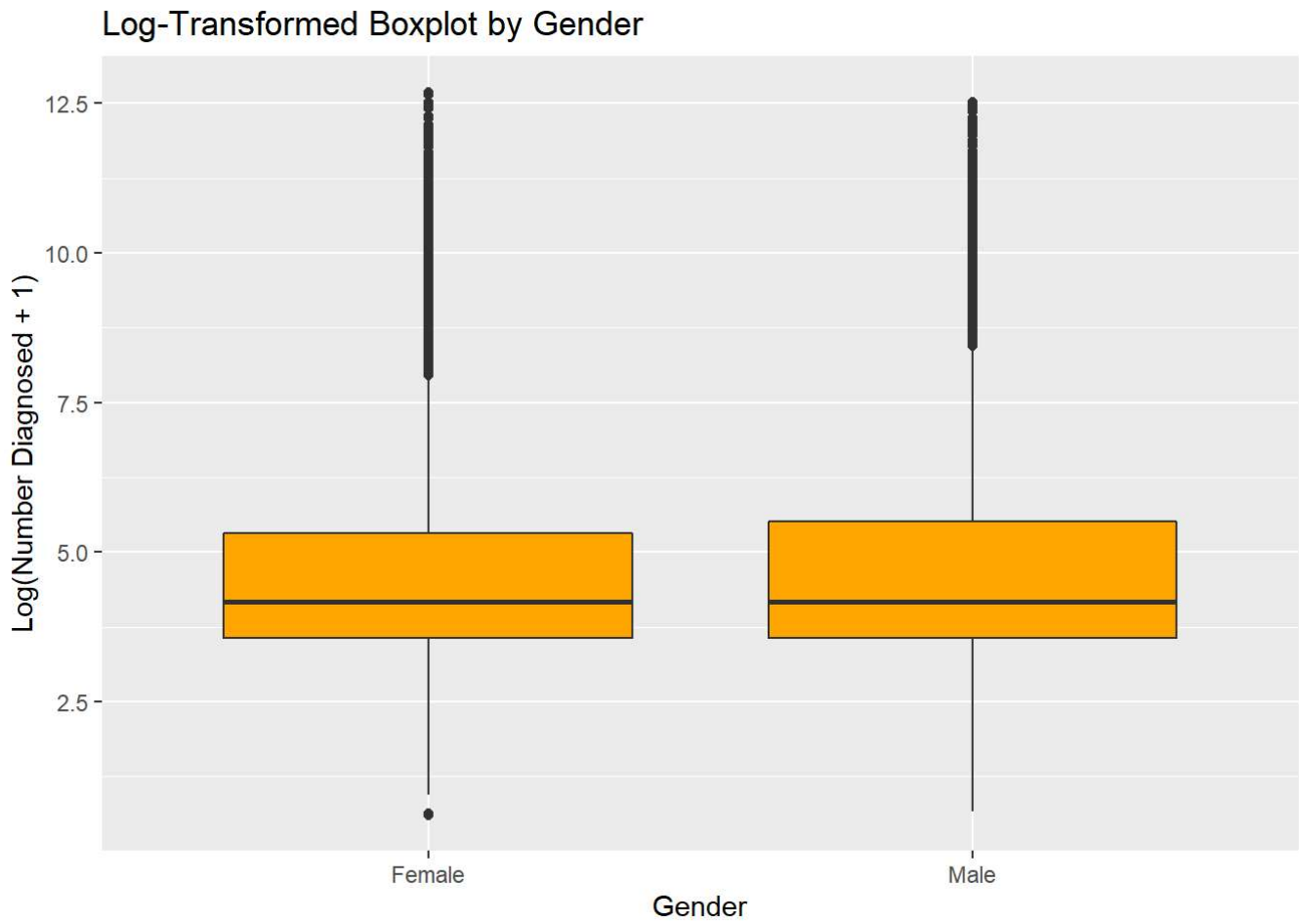


Insights

1. Right-skewed Distribution: Both gender groups (Female and Male) have a large number of extreme outliers at the top end. Most of the data points are clustered near 0, meaning that in most counties (or units of measurement), the number diagnosed is relatively low.
2. Similar Central Tendencies: The median line (inside the boxes) for both genders appears to be very close to the bottom, suggesting similar central tendencies (e.g., median values) for diagnosis counts in both groups.
3. Presence of High Outliers: The presence of many high outliers in both genders may indicate a small number of areas with very high diagnosis counts, possibly corresponding to densely populated counties or high-incidence regions.

Diagnosed Cases by Gender with log transformation

```
ggplot(gender, aes(x = Stratification1, y = log1p(Number_Diagnosed))) +
  geom_boxplot(fill = "orange") +
  labs(title = "Log-Transformed Boxplot by Gender", x = "Gender", y = "Log(Number Diagnosed +
1)")
```



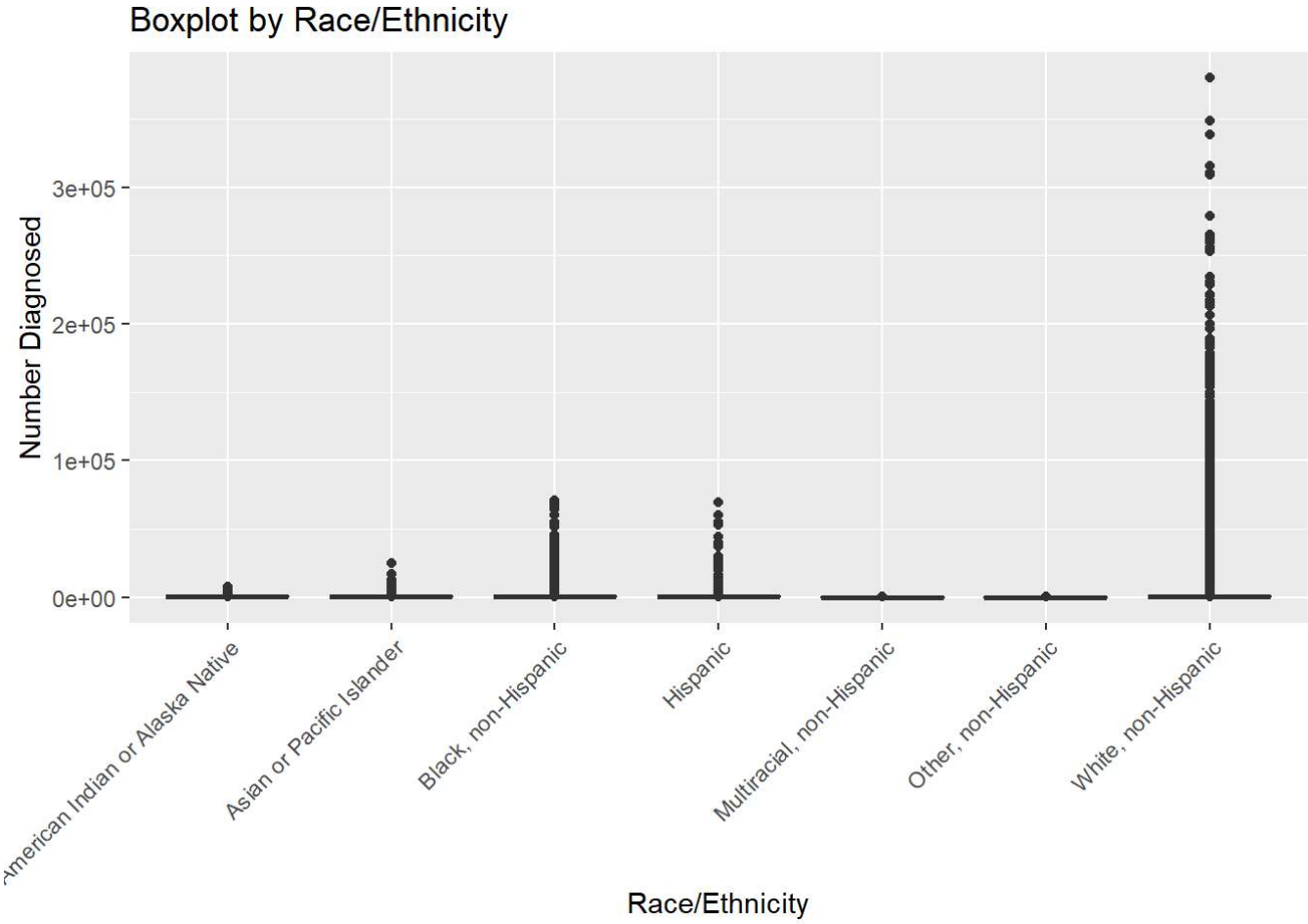
Insights

1. Comparable Distributions: Both Female and Male groups show very similar distributions after log transformation. The median values (central horizontal lines in the boxes) and interquartile ranges (IQRs) are nearly identical.
2. Presence of Outliers: Even after log transformation, both genders still show outliers above the upper whisker, indicating that a few areas still have relatively high diagnosis counts. However, the scale now reveals these outliers more proportionately.
3. Skewness Reduced: The transformation reduces the right skew, making it easier to compare gender groups without distortion from extreme values.
4. No Substantial Gender Gap: There is no significant visual difference in the number diagnosed between males and females based on this visualization.

Diagnosed Cases by Race/Ethnicity

```
race <- final_df %>% filter(StratificationCategory1 == 'Race/Ethnicity')

ggplot(race, aes(x = Stratification1, y = Number_Diagnosed)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot by Race/Ethnicity", x = "Race/Ethnicity", y = "Number Diagnosed")+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

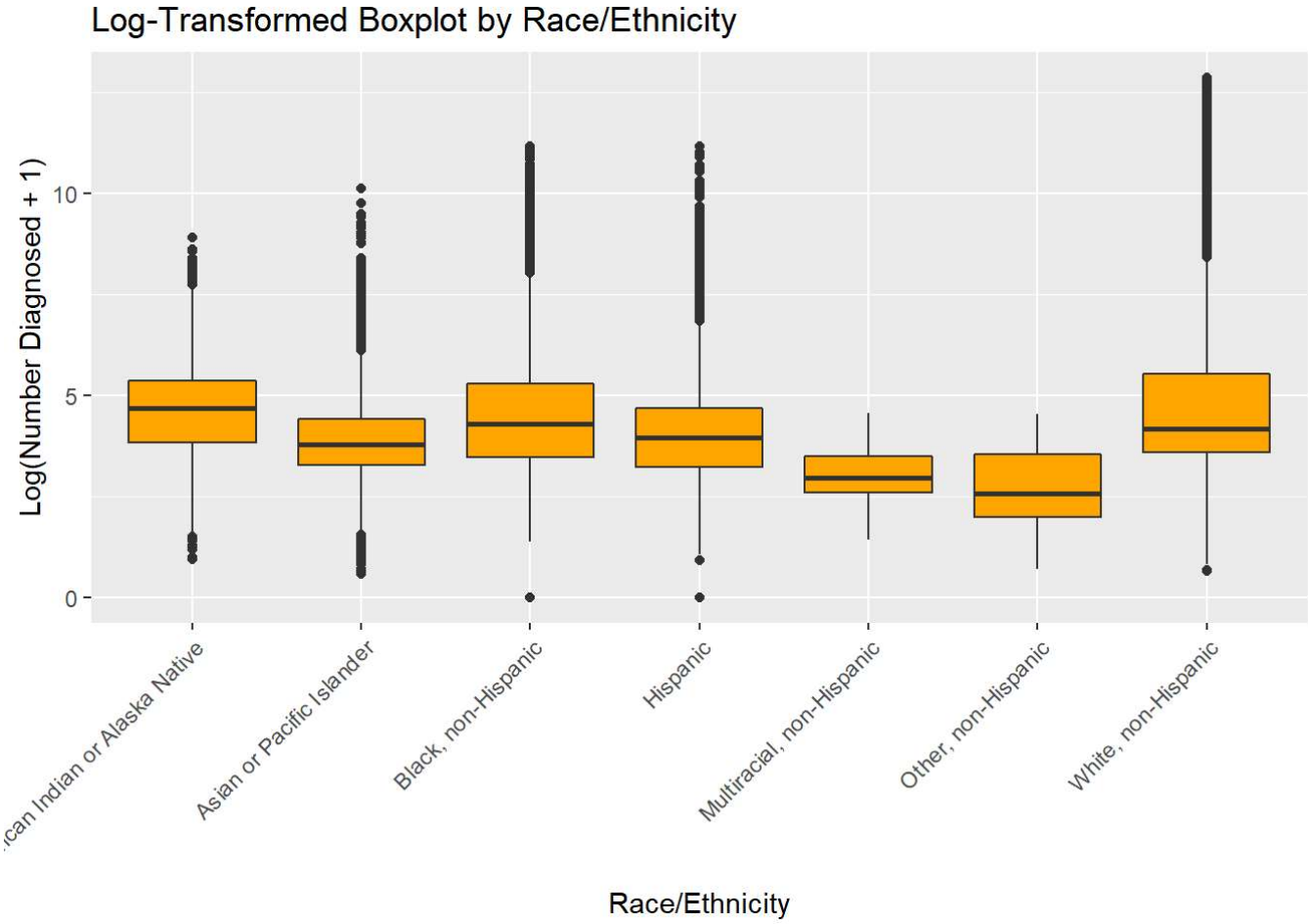


Insights

1. White, non-Hispanic stands out: This group has significantly higher counts and more extreme outliers compared to all other racial/ethnic groups. This may reflect population distribution (i.e., there are simply more White, non-Hispanic individuals in many regions) or other structural factors.
2. Other groups have low medians: Most of the other race/ethnicity categories show low medians and IQRs, with boxplots nearly flattened at the bottom of the y-axis, making them hard to interpret due to scale compression.
3. Right-skewed distribution across all groups: Similar to the gender plot, the raw scale makes the plot less informative due to extreme values dominating the view.

Diagnosed Cases by Race/Ethnicity with log transformaion

```
ggplot(race, aes(x = Stratification1, y = log1p(Number_Diagnosed))) +
  geom_boxplot(fill = "orange") +
  labs(title = "Log-Transformed Boxplot by Race/Ethnicity",
       x = "Race/Ethnicity", y = "Log(Number Diagnosed + 1)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

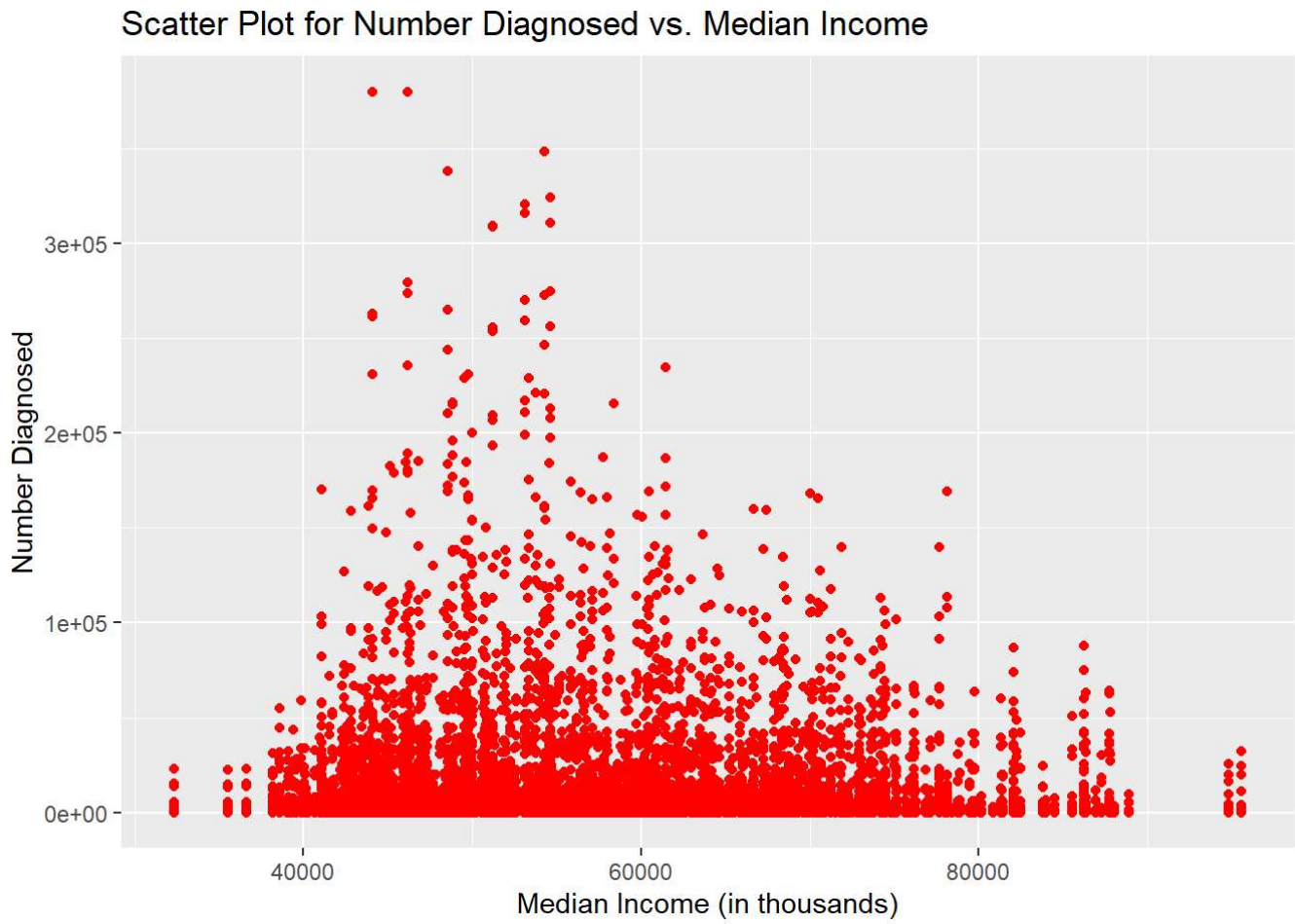


Updated Insights

1. More balanced visualization: Log transformation successfully reduces the skew and compresses extreme values, allowing for a clearer comparison between racial/ethnic groups.
2. White, non-Hispanic still has the highest values: This group retains higher median and upper whisker values, indicating consistently higher diagnosis counts even after adjusting for skew. Suggests either larger population size or potentially higher prevalence/reporting rates in this group.
3. Other groups show similar central tendencies: Groups like Asian or Pacific Islander, Hispanic, and Other, non-Hispanic display similar medians and interquartile ranges, mostly centered around log(4) to log(5)

Diagnosed number vs median income

```
ggplot(final_df, aes(x = Median_Income, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Median Income", x = "Median Income (in tho
usands)", y = "Number Diagnosed")
```

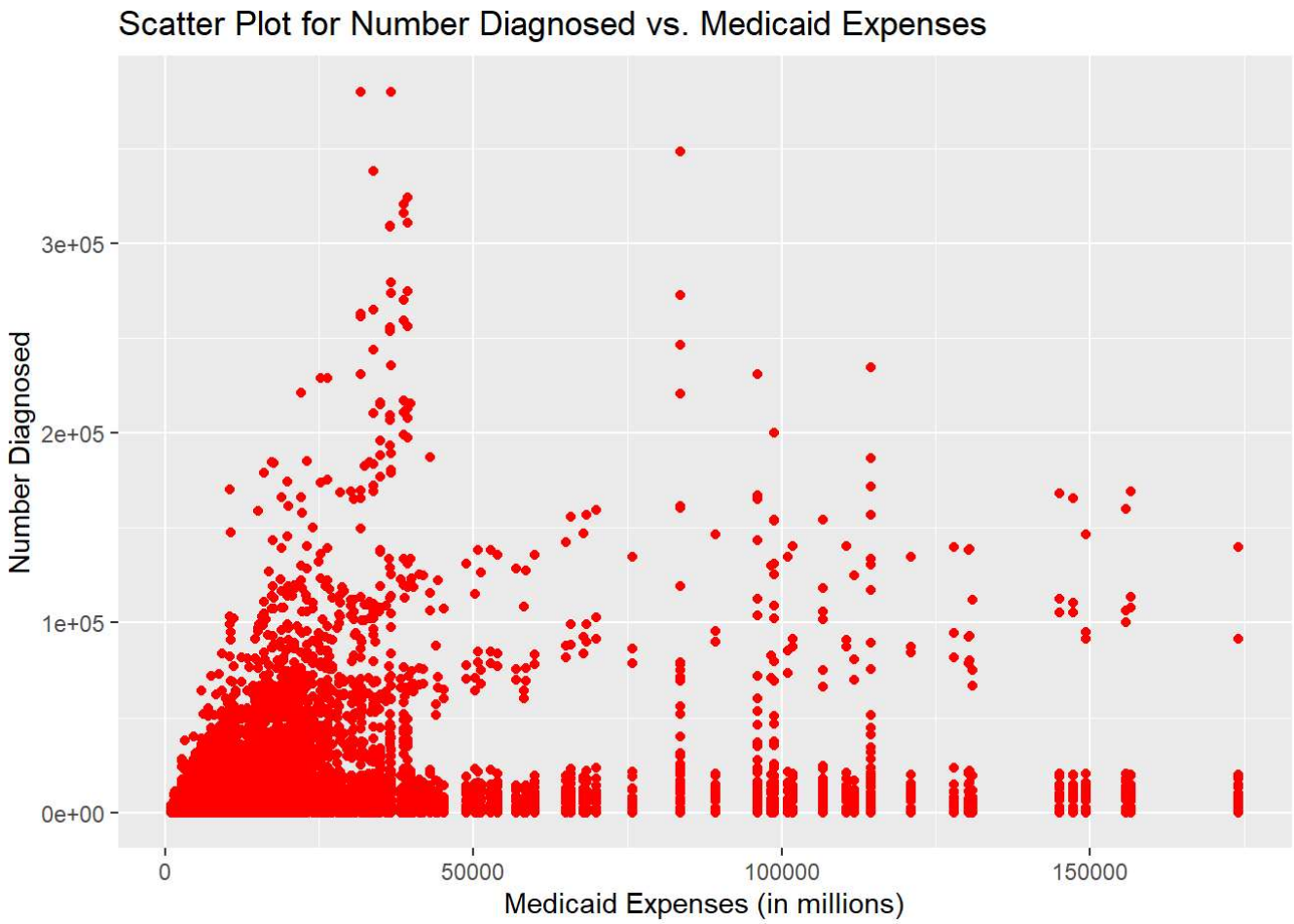


Insights

1. No clear linear trend: The data appears widely scattered with no obvious upward or downward slope, suggesting little to no linear correlation between median income and number diagnosed.
2. High variability at mid-income levels: The most variability in diagnosis counts occurs in the $40,000–$70,000 range. This may reflect areas with larger populations—income alone doesn't explain the variance.
3. Fewer high-count outliers at higher incomes: As median income increases (especially above ~$75,000), we see fewer extreme diagnosis counts. This could hint at lower diagnosis rates or smaller populations in higher-income areas.

Diagnosed number vs medicaid expense

```
ggplot(final_df, aes(x = Medicaid_Expenses, y = Number_Diagnosed)) +
  geom_point(color = "red") +
  labs(title = "Scatter Plot for Number Diagnosed vs. Medicaid Expenses", x = "Medicaid Expenses
(in millions)", y = "Number Diagnosed")
```



Insights

1. No strong linear relationship: Like the previous scatter plot, this one shows no clear linear trend. The points are widely spread across the range of Medicaid expenses.
2. High concentration at lower expenses: A dense cluster of data points lies below $50,000 million in Medicaid expenses. Many of these still show relatively high diagnosis counts. This suggests that higher diagnosis numbers don't necessarily require the highest Medicaid expenditures, possibly due to population size effects.
3. Sparse data at high expense levels: As Medicaid expenses exceed $100,000 million, the number of points decreases and diagnosis counts tend to be lower, or highly variable. These may correspond to states with large Medicaid budgets but low per-case reporting or diagnoses.

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correlation analysis

cor_matrix <- final_df %>% select(where(is.numeric)) %>% cor(use = "complete.obs")

corrplot::corrplot(cor_matrix, method = "circle")